

March 5, 2024

RE: Mr. Rakesh Thoppaen is an Expert in Developing AI Solutions for Diverse Industries.

To Whom It May Concern:

I am Vishnu B Raj, Co-founder and Chief Technical Officer at DeepFlow Technologies, with many years of experience in machine learning and computer vision. During my time here, I have had the chance to lead and teach teams, greatly helping our area of work. Also, I am a Ph.D. scholar in Data Science from the Indian Institute of Technology Palakkad, along with awards like the Siemens Make IT Real Hackathon 2019 and the Reboot Kerala Hackathon 2020. Also, I have written articles, such as a review on Generative Adversarial Networks, showing my dedication to serious study and new ideas. With my wide experience and proven success, I feel confident to talk about the skills and strengths of Mr. Rakesh Thoppaen in the field of Data Science, particularly in machine learning, deep learning, and generative AI.

As the CTO and Co-founder of DeepFlow Technologies, I have been lucky to experience the great guidance given by Mr. Thoppaen. His help has been very useful in moving our company goals forward and improving our team's technical skills. His knowledge and advice have been key in creating a place where we can learn and come up with new ideas.

Our company works on improving farming methods with high-tech solutions. We create AI-based tools to help increase crop production and address farming problems, supporting farmers and encouraging farming that's good for the environment. Our novel ideas are focused on bringing big changes to farming by adding modern technology to old farming ways. A significant aspect of our technological advancement includes the integration of computer vision-based AI models to identify and classify crop diseases and that is where Mr. Thoppaen's mentorship for our company comes in.



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In February 2019, Mr. Thoppaen conducted a comprehensive training program titled "Computer Vision and Deep Learning using Python and OpenCV" at the Government College of Engineering, Kannur. This program, sponsored by the Department of Technical Education, Government of Kerala, focused on critical areas such as image classification, object detection, and semantic segmentation, providing essential skills and knowledge. The training was a 5-day short-term training program on "Computer Vision and Deep Learning using Python and OpenCV," organized by the Department of Computer Science and Engineering, and sponsored by the Department of Technical Education, Government of Kerala. The training aimed to upskill faculties and research students working in AI, covering basics of computer vision, deep learning, data collection techniques, artificial neural networks, convolution neural networks, data augmentation, object detection, semantic segmentation, and generative adversarial networks, etc. Apart from that, it helped the audience gain perspective on the practical application of AI in the industry and the ethical considerations while implementing an AI solution at scale.

During Mr. Thoppaen's training, his skill and guidance stood out as he led our team through the details of AI and computer vision, especially in recognizing and categorizing crop diseases. His way of explaining complex ideas in a simple way was very helpful which made a large impact on our staff to this day.

Furthermore, Mr. Thoppaen's knowledge has greatly changed DeepFlow Technologies. His understanding of how AI can be used in farming has helped us improve our AI models significantly. This has led to better ways to identify and handle crop diseases. This improvement has not only made our platform better but also helped the farming community by giving them the tools they need for improved crop care and higher yields.

With extensive experience in AI, machine learning, and computer vision, particularly the application of these technologies to diverse industries such as semiconductor, insurance and pharmaceuticals, Mr. Thoppaen is a tremendous asset to U.S. industry. His adept skills in technical architecture, including successful implementations of data-driven solutions such as Generative AI and machine learning models, will drive technological advancements and efficiencies across U.S. industries. His creation of AI solutions for complex industry problems combined with AI technical team leadership, fostered through the development and mentoring of technical teams, drives innovation and operational efficiencies across diverse industry sectors.

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In conclusion, I truly believe that Mr. Rakesh Thoppaen's outstanding skills and creative methods in Data Science, especially in Artificial Intelligence, machine learning, and deep learning, make him extremely valuable to the tech and industrial areas of the United States. His dedication to improving AI use in different fields is not only praiseworthy but also crucial for the ongoing improvement and growth of these areas. Therefore, I highly recommend Mr. Thoppaen for the National Interest Waiver petition, as his work is likely to bring great benefits to the U.S. by pushing technology forward and solving problems.

Sincerely,



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